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Salem (Dt)

QSS **QUARTERLY EXAMINATION - 2025**

12 - STD **CHEMISTRY**

Time : 3.00 Hrs Marks : 70

I Note : Answer all the questions. Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer. 15 X 1 = 15

1. Graphite and diamond are
   a) covalent and molecular cyrstals b) ionic and covalent crystals
   c) both covalent crystals d) both molecular crystals
2. The yellow colour in Nacl crystal is due to
   a) excitation of electrons in F centers b) reflection of light from Cl ion on the surface
   c) refraction of light from Na^+ ion d) all the above
3. Dry distillation of calcium formate gives
   a) Ethanal b) Ethanol c) Proponone d) Methanal
4. $CH_2 = CH_2 \xrightarrow[ii)\ Zn/H_2O]{i)\ O_3} X \xrightarrow{NH_3} Y$ Y is
   a) formaldehyde b) diacetone ammonia
   c) hexamethylene tetramine d) oxime
5. Which of the metal is extracted by Hall - Heroult process?
   a) Al b) Ni c) Cu d) Zn
6. Which one of the following ores is best concentrated by froth - floatation method?
   a) Magnetite b) Hematite c) Galena d) Cassiterite
7. Which of the following is not Sp^2 hybridised?
   a) Graphite b) Graphene c) Fullerene d) Dry ice
8. XeF_6 on complete hydrolysis gives
   a) $XeOF_4$ b) $XeOF_2$ c) XeO_3 d) XeO_2
9. The most common oxidation state of actinoids is
   a) + 2 b) + 3 c) + 4 d) + 6
10. Which of the following is used as the source of gamma rays?
    a) Xe b) Ar c) Rn d) Kr
11. The rate constant of a reaction is 5.8 X 10^{-2} S^{-1}. The order of the reaction is
    a) First order b) zero order c) second order d) third order
12. Which of the following fluoro compounds is most likely to behave as a Lewis base?
    a) BF_3 b) PF_3 c) CF_4 d) SiF_4
13. The relationship between the solubility product (Ksp) and molar solubility (s) for $Ag_2(CrO_4)$ is a) $Ksp = S^3$ b) $Ksp = S^2$ c) $Ksp = 4S^3$ d) $Ksp = 3S^2$
14. $HOCH_2 - CH_2OH$ on heating with periodic acid gives
    a) Methanoic acid b) Glyoxal c) Metyhanol d) CO_2

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15. Williamson synthesis of preparing dimethyl ether is a/ an
a) SN^1 reaction b) SN^2 reaction
c) electrophilic addition d) electrophilic substitution

II Note : Answer any 6 questions. Question NO. 24 is compulsory. 6 X 2 = 12

16. Write acid leaching with an example.
17. Write a note on hydroboration.
18. What is the hybridisation of iodine in IF_7. Give its structure.
19. What are interstitial compounds?
20. What are point defects?
21. Define Raw law.
22. What is common ion effect?
23. Write Schotten - Baumann reaction.
24. What is Tollens reagent and Fehlings solution?

III Note : Answer any 6 questions. Q. NO. 33 is compulsory. 6 X 3 = 18

25. Write the differences between mineral and ore.
26. Write the conditions for Catenation.
27. Sulphuric acid is dibasic - why?
28. Why d block elements form co -ordination complexes.
29. What is meant by co ordination number? What is the co -ordination number of atoms in a bcc structure?
30. Write Reimer - Tieman reaction.
31. What is Urotrophine? Write the uses and give the structure.
32. What are buffer solution. Give example.
33. A zero order reaction is 20% complete in 20 . minutes. Calculate the value of the rate constant. In what time will the reaction be 80% complete?

IV Answer all the questions. 5 X 5 = 25

34. a) Explain magnetic separation method. **(OR)**
b) i) CO is a reducing agent. Justify with an example. ii) Write the uses of Boric acid.
35. a) i) Write the reduction property of phossphine. ii) Write the reason for the anamolous behaviour of nitrogen. ***(OR)*** b) What is Lanthanide contraction? Explain its effects.
36. a) Differentiate cystalline and amorphous solids. ***(OR)*** b) i) Give two examples for first order reaction. ii) Differentiate - order of reaction and molecularity of a reaction.
37. a) Derive an expression for Oswald's dilution law. ***(OR)***
b) Explain the mechanism of Cannizaro reaction.
38. a) i?) Write Van - Arkel process. ii) How bleaching powder in prepared?. ***(OR)***
b) An organic compound C_2H_6O (A) heated with Con H_2SO_4 at 443 K to give an unsaturated hydrocarbon C_2H_4(B) . Which on treatment with Bayer's reagent to give compound $C_2H_6O_2$(C) Which is used an antifreeze in automobile radiator. Compound (C) distilled with con. H_2SO_4 to give cyclic compound $C_4H_8O_2$ (D) . Identify A,B, C and D. Write the equations.

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